

4. ERD

Based on the business operations of Cocoon, the following entities were identified to support customer management, product sales, order processing, staff operations, and payment management.

1. Entity: Customers

Attribute	Data Type	Key	Description
customer_id	INT	PK	Unique customer identifier
full_name	VARCHAR(100)		Customer full name
phone_number	VARCHAR(20)		Contact number
email	VARCHAR(100)		Email address
gender	VARCHAR(10)		Gender
date_of_birth	DATE		Date of birth
address	TEXT		Customer address
loyalty_points	INT		Reward points

2. Entity: Categories

Attribute	Data Type	Key	Description
category_id	INT	PK	Unique category identifier
category_name	VARCHAR(100)		Product category name

3. Entity: Products

Attribute	Data Type	Key	Description
product_id	INT	PK	Unique product identifier
product_name	VARCHAR(150)		Product name
category_id	INT		Product category reference

price	DECIMAL(10,2)	Selling price
stock_quantity	INT	Available stock quantity
description	TEXT	Product description
status	VARCHAR(20)	Available / Out of Stock

4. Entity: Departments

Attribute	Data Type	Key	Description
department_id	INT	PK	Unique department identifier
department_name	VARCHAR(100)		Department name

5. Entity: Staff

Attribute	Data Type	Key	Description
staff_id	INT	PK	Unique staff identifier
full_name	VARCHAR(100)		Staff full name
phone_number	VARCHAR(20)		Contact number
email	VARCHAR(100)		Staff email
position	VARCHAR(50)		Job title
department_id	INT		Department reference

6. Entity: Orders

Attribute	Data Type	Key	Description
order_id	INT	PK	Unique order identifier
customer_id	INT		Customer reference
staff_id	INT		Staff reference
order_date	DATETIME		Date and time of order
sales_channel	VARCHAR(50)		Website / Store / Shopee / TikTok Shop

status	VARCHAR(30)	Pending / Completed / Cancelled
total_amount	DECIMAL(10,2)	Total order value

7. Entity: Order_Details

Attribute	Data Type	Key	Description
order_detail_id	INT	PK	Unique order detail identifier
order_id	INT		Order reference
product_id	INT		Product reference
quantity	INT		Quantity purchased
unit_price	DECIMAL(10,2)		Price per item
subtotal	DECIMAL(10,2)		Total amount per item

8. Entity: Payments

Attribute	Data Type	Key	Description
payment_id	INT	PK	Unique payment identifier
order_id	INT		Order reference
payment_method	VARCHAR(50)		Cash / Card / Transfer / E-wallet
payment_date	DATETIME		Payment date
amount	DECIMAL(10,2)		Paid amount
payment_status	VARCHAR(20)		Paid / Unpaid / Refunded

9. Entity: Promotions

Attribute	Data Type	Key	Description
promotion_id	INT	PK	Unique promotion identifier
promotion_name	VARCHAR(100)		Promotion campaign name

discount_rate	DECIMAL(5,2)	Discount percentage
start_date	DATE	Promotion start date
end_date	DATE	Promotion end date

Summary of Primary Keys

- customer_id
- category_id
- product_id
- department_id
- staff_id
- order_id
- order_detail_id
- payment_id
- promotion_id

Entities:

- Customers
- Categories
- Products
- Departments
- Staff
- Orders
- Order_Details
- Payments
- Promotions

